

Types of Injuries Presenting to the Emergency Department Under the Impact of Hurricane Michael in Florida in 2018

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Abstract

Extreme weather events such as hurricanes are becoming increasingly frequent and severe, contributing to a rising burden of injuries and emergency department (ED) visits. Florida, heavily affected by Hurricane Michael in 2018, provides a critical setting for examining injury patterns and healthcare utilization following a major meteorological disaster. This retrospective observational study analyzed 3522 injury-related ED visits from October to December 2018 in the five most impacted Florida counties, using data from the HCUP State Emergency Department Database (SEDD). We characterized the distribution of injury types and examined differences across demographic groups, as well as the relationship between insurance coverage and ED disposition.

Abrasions and contusions were the most common injuries (56.9%), followed by mental health related diagnoses (35.7%). Children under 18 years old had the highest proportion of abrasion/contusion injuries, while adults over 65 had the greatest burden of fractures. Hispanic individuals had the lowest proportion of mental health diagnoses. Across all insurance types, most patients were discharged home; however, Medicare beneficiaries had the highest likelihood of transfer or hospital admission. Multinomial logistic regression showed that older age groups and more severe injury categories (fractures, mental health disorders, puncture wounds) significantly increased the odds of transfer relative to routine discharge, while sex and race/ethnicity were not significant predictors.

Introduction

Background

As climate change becomes more severe each year, extreme weather events such as hurricanes, floods, and extreme heat are leading to increased health risks, including abrasion, puncture wound, fracture, traumatic infections and mental health disorders. These conditions often result in higher volume of emergency room visits. Emergency departments (EDs) often serve as the first point of medical contact for individuals injured during severe weather events, making them an ideal setting to investigate the association between injury patterns and extreme weather. Florida, which is frequently impacted by hurricanes and severe storms, serves as a critical setting for studying these effects. Hurricane Michael was one of the most powerful hurricanes to strike the southeastern United States in recent decades, and its impact provided a unique opportunity to examine how extreme weather events influence injury occurrence and healthcare utilization. With the aging population, substantial racial and ethnic diversity, and the variety of insurance types, Florida can offer valuable insights into how demographic characteristics are associated with the pattern and outcome of injuries presented in ED. Understanding these patterns can

offer valuable insights into the public health impact of hurricanes and help guide future preparedness and response strategies to reduce injury burden during extreme weather events.

Gaps in the literature

Previous studies have typically investigated single-injury related ED visits (alcohol-drinking, blunt chest, spin injury, etc.), or focused on subgroups of the population (such as children and women,). In contrast, research on injuries associated with extreme weather events is limited. Furthermore, much of the literature has not sufficiently explored how injury types vary across multiple dimensions such as age, sex, race/ethnicity, and how insurance type influences hospitalization rate and treatment patients receive. These gaps limit our understanding of how population characteristics may influence the likelihood of presenting with specific types of injuries to ED, and socioeconomic factors, such as insurance coverage, may affect the outcome of the patient, which would eventually lead to re-hospitalization or higher cost of medical bills. State-level databases such as the HCUP State Emergency Department Database has rarely been used to examine ED visits in large and heterogeneous populations like that of Florida, particularly in research focused on severe weather events.

Specific aims and hypotheses

The primary aim of this study is to characterize the distribution of injury types among patients presenting to Florida EDs on October 10, 2018, when Hurricane Michael happened. Specifically: 1) Identify the most common injury categories (e.g., abrasion, puncture wound, etc.) among ED patients; 2) Examine how injury types differ across age, sex, race/ethnicity; 3) Explore whether insurance type affected the outcome of patient (hospitalization, discharge). Here we hypothesize that: 1) certain injury types will be more prevalent among men; 2) Injury patterns may vary across racial and ethnic groups, partly because individuals of the same race or ethnicity often reside in geographically concentrated areas; 3) Disposition of the patient at discharge may vary by insurance type due to differences in coverage, care accessibility, and admission practices across Medicare, Medicaid, and private insurers.

New contribution of this study to the literature

By integrating a broad classification of injury types with demographic and insurance data, this study provides a comprehensive understanding of patterns of emergency department visits after a disaster. This study fills the gap in the literature regarding the lack of research on the demographic characteristics of emergency department visits following meteorological disasters. The results of this study can generate evidence on which populations are most at risk for specific injuries, and help health care providers and organizations anticipate needs and allocate resources more effectively.

Conceptual framework

This study is guided by the idea that injuries are not only influenced by personal factors like age and sex, but also by broader social factors such as race/ethnicity. These characteristics shape how people experience injuries and whether they seek

care in the emergency department. By examining injury types, we can better understand patterns of disparity and identify groups most in need of targeted prevention. Understanding these disparities is also essential for improving emergency preparedness, as communities with higher injury burdens may require additional resources, planning, and support during severe weather events or other crises. Strengthening preparedness efforts with an equity-focused approach can help ensure that emergency response systems are responsive to the needs of the most vulnerable populations.

Method

Overview

This study employed a retrospective observational design to analyze patterns of emergency department (ED) visits across the state of Florida during 2018, with a particular focus on the period following Hurricane Michael (October to December 2018). The study aimed to characterize the types and patterns of injuries observed after the disaster and to determine whether these injury profiles varied across different demographic and socioeconomic groups.

By using statewide emergency department data, this research aims to provide a comprehensive assessment of post-disaster health outcomes and the populations most affected. The analysis focused on identifying the most common categories of injury, exploring differences across age, sex, race/ethnicity, and insurance status, and assessing whether insurance limitations shortened ED stays by leading some patients to leave with only prescriptions instead of receiving extended treatment. This study was exempt from IRB review because it uses de-identified HCUP State Emergency Department data, which do not contain any personally identifiable information.

Data source

Data were obtained from the Healthcare Cost and Utilization Project (HCUP), specifically the Florida State Emergency Department Database (SEDD) for the year 2018.

The SEDD captures visit-level information on all emergency department encounters that do not result in an inpatient admission. It contains detailed patient demographics, diagnostic, payer information, and visit outcomes, allowing comprehensive assessment of injury patterns and associated factors across the state. The use of administrative data enabled a population-based analysis of injury types related to the extreme weather.

Study population

The study included all patients residing in the five most heavily affected counties in Florida (Bay, Gulf, Calhoun, Jackson, and Franklin Counties) from October to December 2018 who presented to the ED with an injury-related diagnosis in their medical record. Injuries were identified using ICD-10-CM codes, including abrasions and contusions (S40, S50, S60, S80, S90, S00), puncture wounds (S31), fractures

(S42, S72), and mental health disorders (F410, F43, R06). Patients without an injury-related diagnosis or those with incomplete demographic information were excluded.

Variables of interest

Outcome variables include: 1) Categorical variables derived from ICD-10-CM codes (abrasion, puncture wound, fracture, traumatic infections, and mental health disorders); 2) Disposition of the patient at discharge (discharged home, transfer or others).

Independent variables consisted of key demographic and socioeconomic factors including: 1) Age groups (0-18, 19-35, 36-65, 65+); 2) Sex (Female, male); 3) race/ethnicity groups (White, Black, Hispanic, Asian and Pacific islanders, others); 4) Insurance type (Medicare, Medicaid, private insurance, self-pay, others).

Furthermore, outcome variables were stratified based on age groups, sex, race groups, and insurance type.

Analytic approach

Descriptive statistics were first used to summarize patient characteristics and the distribution of injury types across subgroups. Stratified analyses were then conducted by key demographic characteristics, including age group, sex, and race/ethnicity. Within each stratum, differences in injury categories were compared to evaluate patterns of disparity.

To address the study aims, we examined: 1) the most common injury types among patients within each demographic subgroup; 2) how the distribution of injury categories (abrasion, puncture wound, fracture, traumatic infection, and mental health disorders) varied across demographic strata; and 3) whether insurance type influenced disposition of the patient at discharge by insurance type.

Pearson's chi-squared tests were used to examine associations between patient disposition and key demographic and clinical variables. Multinomial logistic regression was then performed to identify independent correlates of disposition outcomes. All analyses were performed in RStudio. Statistical significance was defined as $p < 0.05$.

Results

Overview

Table 1 summarizes the descriptive characteristics of the 3,522 visits that presented to the ED after the hurricane in the five most heavily affected counties, based on selected injury-related diagnosis codes. Adults aged 36–65 years accounted for the largest proportion of ED visits (39%). Females were slightly more affected, representing 54.4% of all patients. The majority of patients were White (79.8%), and most were covered by Medicare. More than half of the patients had an estimated annual income below \$45,999. Abrasions or contusions were the most common injury diagnoses (56.9%), followed by mental health–related diagnoses (35.7%). Over 90% of patients were discharged home.

Table 1. Demographic characteristics of patients who presented to the ED after the 2018 hurricane in the five most heavily affected counties in Florida.

Total		3522 (%)
Age Group	0-18	634 (18.0)
	19-35	607 (17.2)
	36-65	1372 (39.0)
	65+	909 (25.8)
Sex	Female	1916 (54.4)
	Male	1606 (45.6)
Race	Asian or Pacific Islander	26 (0.7)
	Black	540 (15.4)
	Hispanic	83 (2.4)
	Other	58 (1.7)
	White	2790 (79.8)
Insurance	Medicare	1150 (32.7)
	Medicaid	728 (20.7)
	Private insurance	841 (23.9)
	Self-pay	517 (14.7)
	No charge	25 (0.7)
	Other	260 (7.4)
Income Group	1 - 45,999	1865 (54.1)
	46,000 - 58,999	1115 (32.4)
	59,000 - 78,999	465 (13.5)
Injury Type	Abrasion/contusion	2004 (56.9)
	Fracture	238 (6.8)
	Mental health disorder	1256 (35.7)
	Puncture wound	24 (0.7)
Disposition of Patients	Discharged Home	3184 (90.4)
	Transferred	285 (8.1)
	Other	53 (1.5)

How injury types differ across age, sex, race/ethnicity

Across age groups, individuals under 18 years old had the highest proportion of abrasion or contusion diagnoses. Adults aged 36–65 and those over 65 exhibited similar proportions of mental health related diagnoses, whereas the oldest group

(65+) had the highest proportion of fractures.

For sex, the distribution of injury types was similar between females and males, with no notable differences observed.

Across the five racial groups, Hispanic patients showed the highest proportions of abrasion/contusion, fracture, and puncture-wound diagnoses but had the lowest proportion of mental health related diagnoses. In contrast, individuals categorized as “Other” (not belonging to the four major racial/ethnic groups) exhibited the highest proportion of mental health related diagnoses.

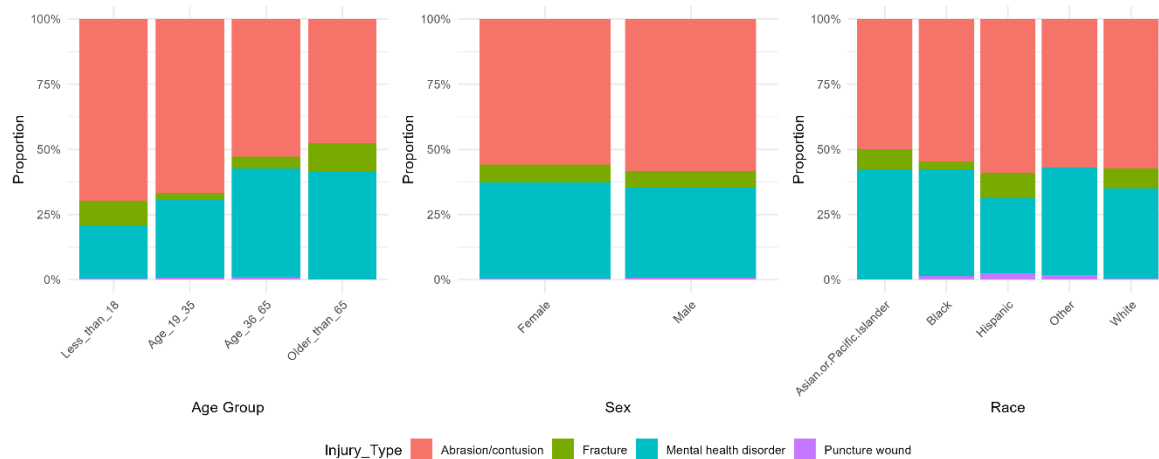


Figure1. Injury type by age, sex, and racial/ethnic groups.

How insurance type affected disposition of patients

As shown in Figure 2, the vast majority of patients in every insurance group (over 80 percent) were discharged home after their ED visit. Individuals covered by Medicaid had the highest rate of routine discharge. However, patients insured by Medicare had the highest proportion of hospitalization or transfers to other types of care facilities.

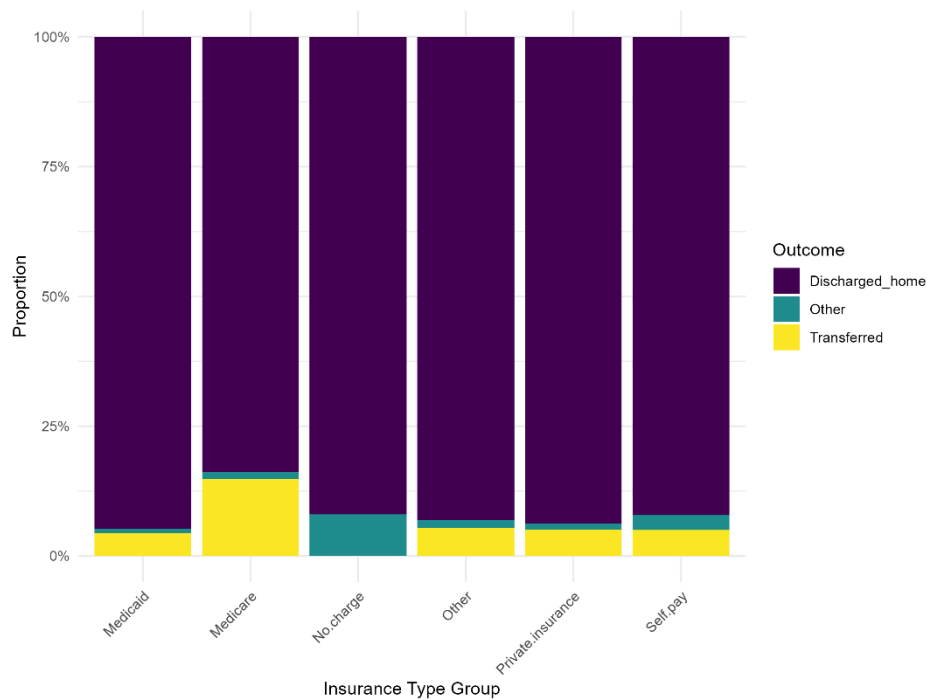


Figure2. Disposition of patients by insurance types.

Variables in relation to disposition of patients

Pearson's Chi-squared test was conducted to examine whether the disposition of patients is associated with age, sex, racial/ethnic groups, and injury types. As Table 2 shows, disposition of patients is strongly associated with age and injury type ($p < 0.001$).

Table 2. Pearson's Chi-squared test for variables in relation to disposition of patients.

	Statistic	p-value	Parameter
Age	121	0.0000	6
Sex	2.82	0.2440	2
Race/ethnicity	3.72	0.8820	8
Injury type	238	0.0000	6

Multinomial logistic regression analysis

To identify correlates of patient disposition, and to evaluate these associations using odds ratios, a multinomial logistic regression model was fitted with age group, sex, race/ethnicity, and injury category as predictors, using discharged home as the reference outcome. For age, the 0-18 group served as the reference. Results indicate that patients aged 36-65 and 65+ have significantly higher odds of being transferred to another hospital or facility, with the greatest increase observed among those 65 years and older. Compared with patients diagnosed with abrasions/contusions, those with fractures, mental health disorders, or puncture wounds exhibit substantially higher odds of transfer. No significant differences were

observed across sex or racial/ethnic groups.

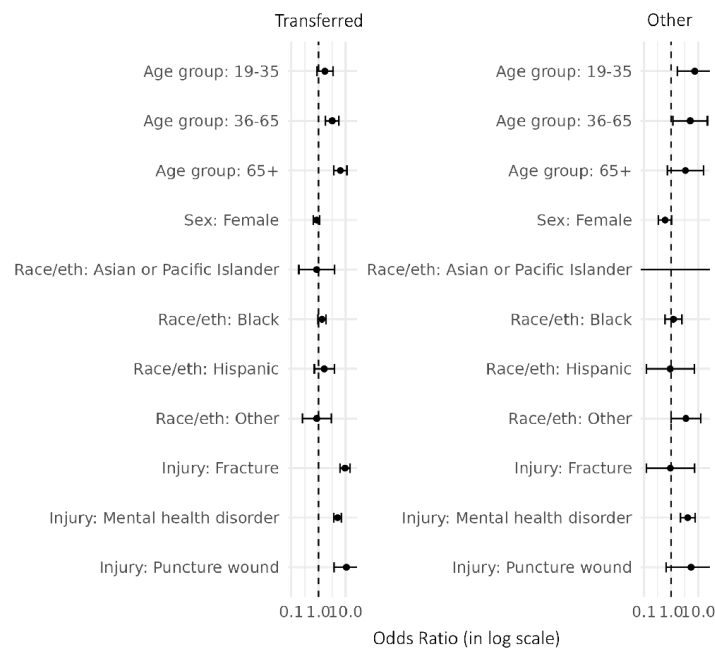


Figure 3. Adjusted odds ratios (log-scale) from multinomial logistic regression examining predictors of patient disposition. The reference outcome category is discharged home. For age groups, the 0–18 years group serves as the reference; for sex, male is the reference; for race/ethnicity, White is the reference; and for injury type, abrasion/contusion is the reference category. Each point represents the odds ratio comparing the likelihood of being transferred or classified as other disposition relative to being discharged home, with 95% confidence intervals displayed on a logarithmic scale.

Discussion

This study examined how demographic characteristics, injury types, and insurance coverage relate to ED disposition patterns among patients presenting with injury-related diagnoses.

The most common injury across all demographic groups in the fourth quarter of 2018 in Florida was abrasion/contusion. The highest proportion occurred among individuals under 18 years old, with the prevalence decreasing steadily with age. This pattern may reflect that children spend more time outdoors and tend to take less caution while walking or running. The second most common category was mental health related diagnoses, which were most prevalent among adults over 35 years old. Prior disaster research has shown that middle-aged and older adults often experience a greater burden of mental health conditions than younger adults following events such as earthquakes, hurricanes, or terrorist attacks, particularly during medium- and long-term follow-up. (delayed ptsd / ptsd after disaster) An interesting finding is that Hispanic individuals had the lowest proportion of mental health–related diagnoses. This pattern may reflect the underutilization of mental

health services among Hispanic populations in the U.S. And many individuals seek support from clergy, family members, or friends. Mental health related stigma has also been widely reported within this community, which may further reduce formal diagnosis. (Latino mental health care) The proportion of abrasion/contusion diagnoses was slightly higher among males, whereas mental health related diagnoses were slightly more common among females; however, these gender differences were small, and overall injury patterns did not differ substantially between males and females.

Insurance status also showed meaningful associations with patient disposition. Across all insurance types, most patients were discharged home, but Medicare beneficiaries had the highest proportion of transfers or hospital admissions. This indicates that patients covered by Medicare tend to be older and more medically complex.

Both the chi-square analysis and the multinomial logistic regression converged on the same conclusion: age and injury type are the strongest correlates of disposition from the ED. Older adults, especially those aged 65 and above, were significantly more likely to be transferred to a hospital or other facilities than younger patients. Injury severity also played a critical role: fractures, mental health crises, and puncture wounds were consistently associated with higher transfer odds compared with abrasion/contusion. On the contrary, sex and race/ethnicity were not significant predictors of disposition.

Strength

These findings highlight important demographic and socioeconomic patterns in ED utilization after a major hurricane. Understanding these disparities can help inform targeted preparedness strategies, resource allocation, and post-disaster healthcare planning to better support vulnerable populations.

Limitation

There are several limitations that should be considered when interpreting these findings. First, the definition of diagnosis codes used in this analysis was relatively narrow. A broader approach would have captured more visits and expanded the analytic population, such as summarizing and including the top ten diagnosis codes during the study period. Additionally, the injury categories employed were broad and may not fully reflect the range of clinical severity or comorbid conditions that influence disposition decisions. Second, the analysis did not include a comparison with the period preceding the hurricane. Comparing the fourth quarter (post-hurricane) to the third quarter could have highlighted changes in the distribution of injury types across populations and provided stronger contextual interpretation of the findings.